Toward a Formal Statement of the No-Refolding Constraint in the Hilbert Substrate Framework

Ben Bray

March 20, 2026

Abstract

This note proposes a formal interpretation of the no-refolding constraint in the Hilbert Substrate Framework (HSF). The central claim is that no-refolding should not be understood merely as a ban on arbitrary graph rewiring or arbitrary factor relabeling. Rather, it is a protection principle for collective constraint-relief structures: once the substrate forms a mesoscopic support pattern that efficiently carries local constraint stress, that organization becomes physically committed. Subsequent dynamics may strengthen, weaken, or lawfully re-express the structure, but may not arbitrarily erase or refactor it without paying a real dynamical cost. The note distinguishes a general no-refolding principle from possible special realizations, including fermionic Jordan–Wigner string structure and topological-like protection, and concludes with candidate operational witnesses for future derivation work.

1 Why this note is needed

The no-refolding constraint has so far been used as a necessary idea in HSF, but it has remained underformalized. It has often been invoked to express the intuition that once structure exists, the substrate cannot freely "refactor" or "rewire" itself as though that structure were arbitrary. However, such phrasing is too loose to support a derived dynamical law.

The present note is an attempt to state the idea more sharply.

The motivating problem is this:

> If the substrate is permanently generative and permanently compressive, then why do some mesoscopic structures persist? Why can the substrate not simply reconfigure itself freely at each step?

The proposed answer is:

> Because some collective structures are not arbitrary arrangements of support. They are dynamically selected solutions to local constraint stress. Once such a solution forms and begins carrying that stress efficiently, its organization is physically committed. Arbitrary destruction or rewriting of that organization is not free.

This is the intended content of no-refolding in the present note.

2 Core intuition

The best intuitive picture is not that no-refolding forbids all rearrangement. Rather, it says that once the substrate finds a working collective pattern for carrying local stress, it cannot pretend that the pattern was arbitrary.

A useful analogy is crystallization in a supersaturated medium. If one has a boiling pot of sugar water and a string is introduced, the string can act as a nucleation scaffold. Once a crystal begins forming around it, the resulting crystal is a real collective structure. One may still destroy it, but one may not simply relabel it away. Undoing it requires actually breaking the organized pattern that has formed.

The proposed HSF analogy is:

- the substrate is the medium,
- the four HSF constraints generate local stress or frustration,
- support patches and links are the explicit structures available to carry that stress,
- a mesoscopic organizer is a collectively stabilized relief pattern,
- no-refolding is the principle that such a relief pattern, once formed, becomes dynamically committed.

3 Constraint stress and collective relief

3.1 Local stress picture

The HSF picture already suggests that the substrate is simultaneously subject to multiple pressures:

- finite bandwidth,
- no-signaling / bounded spread,
- no-forgetting,
- no-refolding itself as a persistence/commitment pressure.

These pressures need not all point in the same direction. In general they are frustrating rather than harmonizing. A local region of the substrate is therefore expected to carry a nontrivial *constraint stress* or *constraint frustration load*.

We do not yet assume that this stress has a single scalar representation. Indeed, it is more natural to regard it first as multi-component:

$$\Sigma_R = \left(\Sigma_R^{(B)}, \Sigma_R^{(S)}, \Sigma_R^{(F)}, \Sigma_R^{(\mathrm{comm})}\right), \tag{1}$$

where:

$$\begin{aligned}
\Sigma_R^{(B)} &: \text{bandwidth load,}\\
\Sigma_R^{(S)} &: \text{spread / signaling load,}\\
\Sigma_R^{(F)} &: \text{no-forgetting load,}\\
\Sigma_R^{(\mathrm{comm})} &: \text{commitment / refolding load.}
\end{aligned}$$

A mesoscopic object or organizer is then interpreted as a support pattern that carries this load more effectively than neighboring alternatives.

3.2 Collective relief

A support pattern should be called a *collective relief structure* if:

1. it is not reducible to a single local site identity,
2. it reduces the local constraint load more effectively than nearby alternatives,
3. its function depends on the coordinated organization of several supports and interfaces.

This language is chosen carefully. The idea is not that the object is metaphysically fundamental. The idea is that it is dynamically favored because it is carrying part of the substrate's local burden.

4 Proposed statement of no-refolding

4.1 Plain-language statement

The plain-language version is:

> Once the substrate forms a collective support pattern that efficiently relieves local constraint stress, that pattern becomes physically committed. It may later be strengthened, weakened, or lawfully re-expressed elsewhere, but it may not be arbitrarily erased or refactored without paying a corresponding dynamical cost.

4.2 Formal draft statement

We now propose the following draft formulation.

> **No-refolding constraint (draft).** Let R be a local region of the substrate, and let O_R be a mesoscopic support structure in that region. If O_R is a stable or metastable collective relief of local constraint stress, then the organization of O_R is physically committed. Any move that destroys, rewrites, or refactors O_R without preserving its relief function through lawful re-expression must incur a corresponding dynamical cost.

This wording does not make no-refolding an absolute ban on change. Rather, it makes it a penalty or barrier against arbitrary undoing of working collective solutions.

5 Lawful re-expression versus unlawful refolding

This distinction is central.

5.1 Lawful re-expression

A move should count as *lawful re-expression* if it:

1. preserves the relief function of the original structure, either in place or by transferring it elsewhere,
2. preserves enough of the relevant relational/bookkeeping structure to maintain physical continuity,
3. does not destroy the collective organization without replacement.

Examples might include:

- demotion of a support patch when its relational role is preserved elsewhere,
- strengthening one interface while weakening another, provided the same collective object remains functionally intact,
- re-expression of support in a new explicit patch that better carries the same local load.

5.2 Unlawful refolding

A move should count as *unlawful refolding* if it:

1. removes or rewrites a currently effective collective relief pattern as though it were arbitrary,
2. destroys the pattern's bookkeeping structure without preserving its function elsewhere,
3. treats a historically and dynamically committed organization as if it were merely a relabeling choice.

Thus no-refolding is not a doctrine of immobility. It is a doctrine that working collective structure is not freely disposable.

6 What exactly is protected?

This is still the main unresolved conceptual question. Several candidates are possible.

6.1 Candidate A: factorization structure

One possibility is that no-refolding protects a particular factorization of the substrate into effective supports. On this reading, once a support decomposition has become physically meaningful, one may not arbitrarily re-factor it.

This captures part of the intended idea, but it is probably too abstract by itself. A factorization is not yet a concrete mesoscopic object.

6.2 Candidate B: interface / link commitment

Another possibility is that no-refolding protects committed interfaces. On this reading, once links or interfaces are actually carrying relational load, they may not be freely rewired.

This is closer, but may still be too narrow. A mesoscopic organizer may be more than the sum of its individual committed links.

6.3 Candidate C: ordering or bookkeeping structure

A third possibility is that what matters is a nonlocal bookkeeping structure: some ordering, parity, sign, or string data that becomes committed once excitations exist.

This is particularly relevant if fermionic sectors are fundamental or if Jordan–Wigner-type strings turn out to be the canonical realization of committed nonlocal bookkeeping.

6.4 Candidate D: collective tension-relief pattern

The strongest present candidate is broader:

> What is protected is the organization of a committed collective solution to local constraint stress.

This includes factorization, interface structure, and bookkeeping structure when those are part of the collective solution, but does not reduce the constraint to any one of them from the start.

This note therefore proposes Candidate D as the best current general statement.

7 General principle versus fermionic specialization

7.1 General no-refolding

The most conservative formulation is:

> No-refolding is a general protection principle for committed collective relief structures.

This allows the framework to host multiple concrete realizations of committed bookkeeping:

- fermionic string/order structure,
- gauge-like interface sectors,
- possibly topological-like pattern protection in some regimes.

7.2 Fermionic specialization

A sharper fermionic version would say:

> In fermionic sectors, the committed bookkeeping protected by no-refolding may appear as Jordan–Wigner-type sign/order structure. Moves that alter support while changing this string structure are not free re-descriptions; they are physically nontrivial and must incur a cost or be disallowed.

This is a strong and attractive specialization. However, the present note does *not* identify it with the entire no-refolding constraint. Instead, it is treated as one important candidate realization.

7.3 Why the distinction matters

If one defines no-refolding immediately as "JW-string preservation," one risks making the whole constraint depend on a particular fermionic representation before the framework has established that all committed structure is fundamentally fermionic.

If instead one says:

> JW-string protection is a concrete fermionic case of a broader no-refolding principle,

then the framework remains general while still allowing the fermionic case to be treated seriously.

This note adopts the second option.

8 Topological-like protection?

The present intuition has some affinity with topological protection, but this must be stated carefully.

The safe claim is:

> No-refolding may have topological-like features insofar as committed collective structures can become history-dependent, path-dependent, and resistant to local rewriting.

However, one should *not* yet call this true topological protection unless one has identified an actual invariant, protected sector, winding, parity class, or other mathematically definite object that remains unchanged under allowed moves.

At present, the evidence supports:

- collective commitment,
- metastability,
- history dependence,
- resistance to arbitrary rewriting.

This is suggestive, but not yet sufficient to claim strict topological protection.

9 Dynamical versus kinematic status

A major unresolved question is whether no-refolding is:

1. a *kinematic restriction* on admissible states/descriptions, or
2. a *dynamical cost principle* that suppresses but does not absolutely forbid certain reorganizations.

The present note favors the second interpretation.

9.1 Reason for favoring a dynamical reading

The sugar-crystal analogy, the mesoscopic-boiling picture, and the language of "paying a cost" all point toward no-refolding being a barrier rather than a strict impossibility.

Under this interpretation:

- lawful re-expression is possible,
- major rewrites are not forbidden in principle,
- but they are disfavored unless they are compensated by an equal or better re-expression of the same relief function.

Thus the present working view is:

> No-refolding is best understood as a dynamical protection principle, not a hard kinematic ban.

10 Candidate operational witnesses

This note does not claim to have derived the final witness. But it does identify what the witness must try to measure.

A future operational witness for no-refolding should detect whether a move:

1. destroys a currently effective collective relief structure,
2. preserves its function through re-expression,
3. or leaves the structure essentially intact.

The most promising current witness families are:

10.1 Function-preservation witnesses

These ask whether the local object still carries the same relief role after the move. Candidate proxies include:

- organizer fidelity under the move,
- preservation of local mutual-information structure,
- preservation of dominant interface activity,
- preservation of shell/core support.

10.2 Commitment-overlap witnesses

These ask whether the move preserves enough of the original object's organization to count as re-expression rather than destruction. Candidate proxies include:

- overlap of pre-move and post-move shell structure,
- overlap of committed interface roles,
- overlap of bookkeeping or string structure in specific sectors.

10.3 Bookkeeping-preservation witnesses

These are especially important if the fermionic/Jordan–Wigner intuition proves central. Candidate future witnesses include:

- preservation of effective ordering data,
- preservation of parity/sign constraints,
- preservation of string-support compatibility.

10.4 Protection-barrier witnesses

These ask whether attempts to remove the object produce a real cost. Candidate proxies include:

- large loss of local constrained-expression score,
- sharp increase in spread or bandwidth burden,
- large no-forgetting penalty under removal.

11 A provisional dynamical statement

The clearest provisional dynamical form is:

> The substrate suppresses moves that destroy a currently effective collective tension-relief structure unless those moves are compensated by an equal or better re-expression of the same relief function elsewhere.

This statement has several advantages:

- it does not freeze the substrate,
- it allows re-expression and churn,
- it distinguishes lawful re-expression from arbitrary refolding,
- it naturally fits a local move principle.

It also pairs well with the broader move-selection picture:

> accept a move only if it improves constrained expression enough to justify its cost.

In that setting, the role of no-refolding is to ensure that the cost of destroying a working collective solution is represented explicitly.

12 What remains unresolved

The present note is a formal clarification, not a finished theorem. The main open questions are:

1. What is the exact protected object: factorization, interface commitment, bookkeeping structure, or some layered combination?
2. Is the fermionic/JW-string case the essence of no-refolding or one special realization?
3. What operational witness best distinguishes lawful re-expression from unlawful refolding?
4. Can one define a genuine invariant or quasi-invariant associated with committed collective relief?
5. Is there a minimal mathematical representation of local constraint stress that makes the relief function precise?

These questions must be answered before no-refolding can be treated as fully derived in a dynamical law.

13 Working conclusion

The present best formulation is:

> No-refolding is a dynamical protection principle for committed collective constraint-relief structures. Once the substrate forms a mesoscopic support pattern that is carrying local stress efficiently, that organization is no longer arbitrary. It may be strengthened, weakened, or lawfully re-expressed, but it may not be freely erased or refactored without paying a corresponding cost.

This general principle is broad enough to include:

- interface commitment,
- factorization persistence,
- fermionic/Jordan–Wigner bookkeeping as a special case,
- and possibly topological-like protection in some regimes.

At present, the most defensible hierarchy is:

1. no-refolding is general,
2. fermionic JW-string preservation is one important candidate realization,
3. topological protection remains a suggestive but not yet established possibility.

One-sentence summary

No-refolding is best understood, at present, not as a ban on all rearrangement, but as the rule that the substrate may not arbitrarily dismantle a working collective solution to its own local constraint stress.